

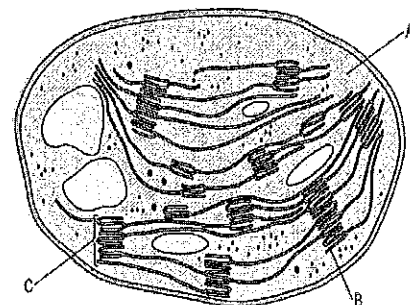
Name: _____

Section: _____

1. Identify the main function of photosystems.
 - A. To absorb and transfer light energy in order to excite electrons
 - B. To generate ATP
 - C. To generate reduced NADP
 - D. To generate oxygen
2. Identify the correct description of a photosystem.
 - A. Arrays of pigments within protein complexes
 - B. Arrays of electrons within membranes
 - C. Arrays of proteins within membranes
 - D. Arrays of electrons within protein complexes
3. During the light-dependent reactions, which component provides the electrons needed to replace those lost from photosystem II?
 - A. Electrons from water molecules
 - B. Oxygen molecules
 - C. Hydrogen ions
 - D. Electrons from photosystem I
4. If six molecules of water were split, determine the number of O_2 , H^+ and e^- that would be formed.
 - A. $3O_2$, $12H^+$, $12e^-$
 - B. $6O_2$, $12H^+$, $12e^-$
 - C. $3O_2$, $6H^+$, $6e^-$
 - D. $6O_2$, $6H^+$, $6e^-$
5. Which of the following conditions would you expect to find within the intermembrane space of the thylakoids while light is being absorbed?
 - A. Low pH conditions due to the high concentration of H^+
 - B. High pH conditions due to the high concentration of H^+
 - C. Neutral pH conditions due to a low concentration of H^+
 - D. Low pH conditions due to a low concentration of H^+

6. The diagram below shows a diagram of a chloroplast.

a. Identify the structures of a chloroplast labelled A-C.

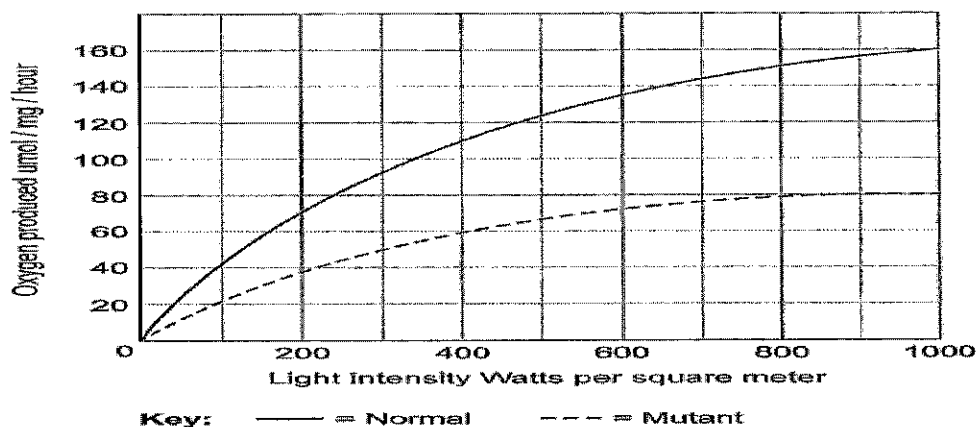


b. Describe the adaptations of structures A and B from the diagram in part a).

Plants can contain more than one type of chlorophyll, a and b. Scientists grew plants that contained a mutant version of chlorophyll a. They investigated the effect of this mutation on the rate of photosynthesis. The scientists:

- Grew the mutant and normal plants in a range of light intensities.
- Isolated the chloroplasts from both types of plants.
- Measured the oxygen produced by the chloroplasts over a period of 20 minutes.

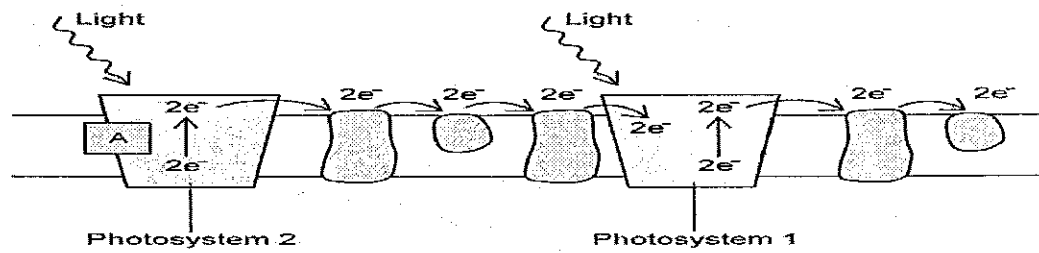
Their results are shown in the graph below:



c. Explain why the scientists used the oxygen produced as a measure of the rate of photosynthesis.

d. Use the graph to calculate the difference in oxygen produced by the chloroplasts from normal plants compared to the mutant plants at a light intensity of 400 Watts/m².

7. The diagram below shows a representation of some of the reactions taking place during the light dependent reactions of photosynthesis.

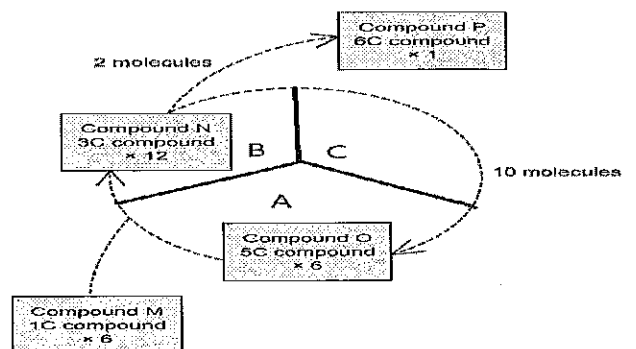


- Name and describe the process that occurs at label A.
- The diagram from part a) shows the loss of electrons from photosystem 2, but does not show how these electrons are replaced. State the source of the electrons that replace those lost at stage A.
- Excited electron in PSII transfer through _____ allowing the _____ to get pumped into the lumen creating _____.
- Name the enzyme that forms ATP. What causes it to function?
- Describe what happens in photosystem I.

8. The diagram below depicts the light-independent reactions of photosynthesis.

- Identify the compounds M, O, and P.

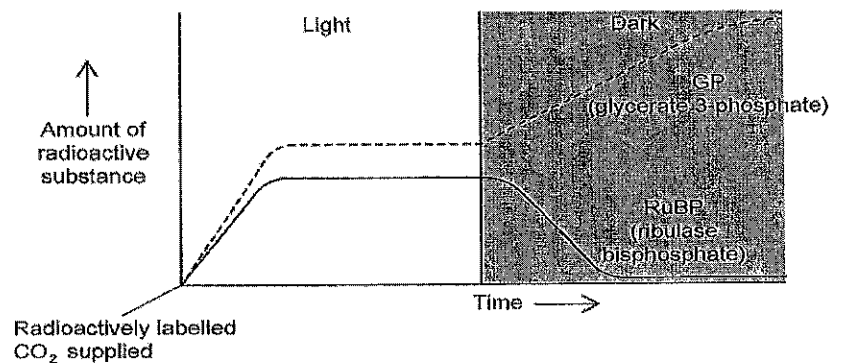
M: _____
 O: _____
 P: _____



- b. Name the enzyme involved in this stage of photosynthesis. _____
- c. Name the energy carriers involved in this stage of photosynthesis. _____
- d. Identify stages A, B, and C. Then describe what happened in each stage.

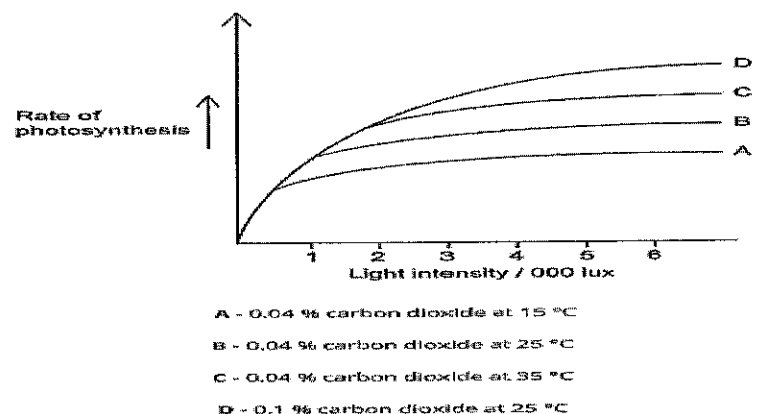
Stage	A	B	C
Name			
Description			

9. An experiment using radioactive carbon was carried out by scientists to investigate the Calvin Cycle. The algae *Chlorella* was exposed to radioactive carbon for different amounts of time. The algae was then analysed for radioactive compounds. The graph below shows the results.



Explain the changes in the amount of radioactive substances in the dark.

10. The graph below shows the effect of light intensity, carbon dioxide, and temperature on the rate of photosynthesis.



temperature on the rate of photosynthesis.

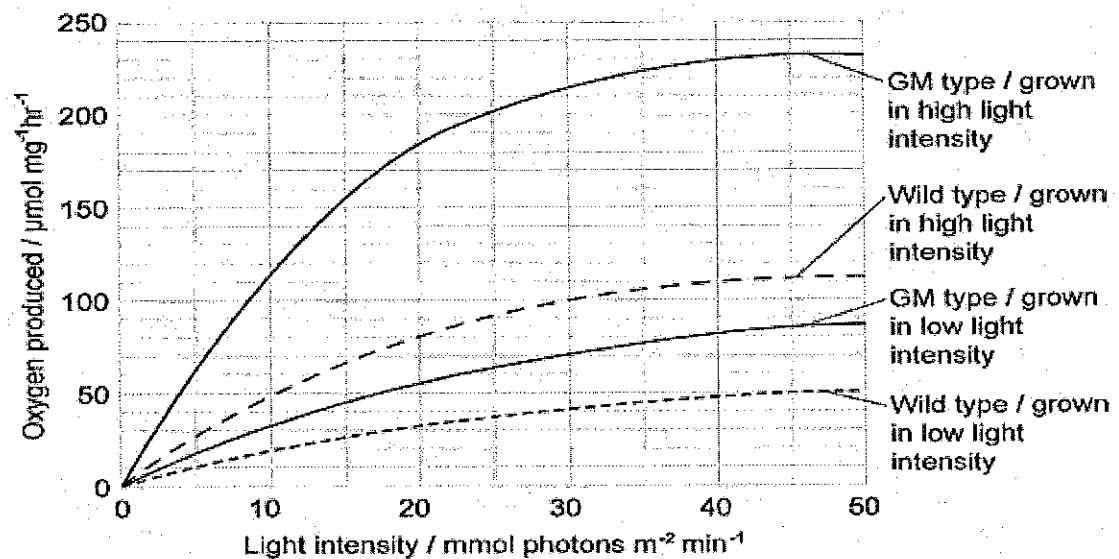
- a. State three factors that could be measured to allow an estimate of the rate of photosynthesis in this experiment.

- b. A commercial farmer growing peppers keeps her greenhouses at 25 °C and circulates air around the greenhouses. Air has a CO concentration of 0.04 % and the ambient light intensity is 3 000 lux. 2

- (i) Use the graph in part a) to state whether the farmer would be more likely to achieve higher pepper yields by raising the carbon dioxide concentration to 0.1 % or by raising the temperature to 35 °C.

- (ii) Explain your answer to part i).

- c. There are two types of chlorophyll in chloroplasts, chlorophyll a and chlorophyll b. Researchers created a genetically-modified (GM) vine plant with an allele that caused them to synthesise higher levels of chlorophyll b than wild-type vine plants. They investigated the effect of this new allele on the rate of plant growth. The researchers grew wild-type and GM vines. They grew some of each in low light intensity and grew others in high light intensity. They extracted chloroplasts from mature plants of both types. Finally, they measured oxygen production at different light intensities by the chloroplasts they had extracted from the plants.



- (i) Oxygen production here is used as a measure of the rate of photosynthesis. State why this is possible.

- (ii) Calculate the percentage improvement in oxygen production caused by the genetic modification for vines grown at high light intensity at an experimental light intensity of 20 mmol photons m⁻² min⁻¹. Give your answer to 3 significant figures.

- d. The researchers suggested that GM plants producing more chlorophyll b would grow faster than wild-type plants in all light intensities. Explain how the data in part c) support this suggestion.
